

Cloud_Computing_Key_Concepts (1)

Short summary | PDF document | 5 pages | css | Generated 3 Jun 2026

AI Summary

Overview

Cloud computing delivers scalable IT services like servers and software over the internet on demand, characterized by resource pooling and rapid elasticity. Key models include IaaS, PaaS, and SaaS, deployed publicly, privately, or hybrid. Virtualization, using VMs and containers, is fundamental.

Why It Matters

DBMS manages structured data, primarily using the relational model. Essential concepts include normalization to reduce redundancy, SQL for data interaction, and ACID properties (Atomicity, Consistency, Isolation, Durability) to ensure reliable transactions and data integrity.

Important Questions

1. Compare and contrast the three main cloud service models (IaaS, PaaS, SaaS) by defining each, providing an example, and outlining the user's responsibility versus the provider's responsibility.
2. Explain the five essential characteristics of cloud computing as defined by NIST, providing a brief description for each. How do these characteristics collectively enable the benefits of cloud adoption?
3. Describe the purpose of normalization in DBMS and explain the first three normal forms (1NF, 2NF, 3NF) with specific conditions for each. Provide an example of a data anomaly that normalization aims to prevent.
4. Define the ACID properties of database transactions. Illustrate how a violation of any one of these properties could lead to data inconsistency or loss in a banking transaction scenario involving a fund transfer.
5. Discuss the concept of virtualization as the foundation of cloud computing. Differentiate between Virtual Machines (VMs) and Containers, highlighting their respective advantages and use cases in a cloud environment.